

Catherine Ge

Prospective doctoral researcher | Digital health, health communication, behaviour change & physical activity

sisfit.cn/CatherineGe | [Research portfolio](#) | [GitHub](#)

RESEARCH PROFILE

Researcher-in-training developing a doctoral profile at the intersection of digital health literacy, health communication, behaviour change and physical activity. Current research preparation combines formal research-design training, systematic evidence synthesis, complex-survey secondary analysis, repeated cross-sectional analysis and device-linked behavioural data.

Digital health literacy	Behaviour change	Physical activity	Health communication
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EDUCATION & FORMAL RESEARCH PREPARATION

Master of Teaching — University of South Australia (UniSA)

GPA 4.46 / 5

PU5027 Fundamentals of Research Design (Online) — University of Aberdeen

2026-27 · In progress · Formal research-design preparation for doctoral study

SELECTED INDEPENDENT RESEARCH PROJECTS

HINTS 7 Secondary Analysis | 2026 · *Complex survey analysis*

- Secondary analysis of the U.S. National Cancer Institute HINTS 7 (2024) public-use dataset.
- Survey-weighted logistic regression using the full-sample weight and 50 JK1 replicate weights; ordinal and model-specification sensitivity analyses.
- Primary adjusted association: OR 1.83 (95% CI 1.42-2.34; n=5,183). Cross-sectional, non-causal interpretation.

[Reproducible project](#)

HINTS 6 + HINTS 7 Repeated Cross-Sectional Analysis | 2026 · *Survey-year interaction*

- Pooled analysis of independent 2022 and 2024 HINTS cycles using the NCI Rizzo merging design with 100 replicate weights.
- Primary survey-year interaction: OR 1.20 (95% CI 0.83-1.73; pooled n=10,838; p=0.330).
- The descriptive gap was larger in 2024, but the formal interaction did not establish a change in the focal association.

[Reproducible project](#)

NHANES Device-Measured Movement Analysis | 2026 · *Device-linked behavioural data / complex survey*

- NHANES 2013-2014 secondary analysis integrating day-level wrist physical-activity-monitor summaries, GPAQ leisure activity, demographics and BMI.
- Primary model used MEC weights, masked strata and PSUs. Device outcome is MIMS movement density, not MVPA minutes.
- Higher self-reported leisure activity was positively associated with device-measured movement density: beta 0.00876 (95% CI 0.00347-0.01404; n=4,726; p=0.003).
- The focal association remained positive across five prespecified movement-data completeness rules and additional model sensitivities.

[Reproducible project](#)

EVIDENCE SYNTHESIS

Two systematic evidence-synthesis projects — In progress

Formal public titles and persistent identifiers are not listed until the source records are fully reconciled. One project has an existing OSF DOI; exact metadata remains pending verification.

METHODS & RESEARCH CAPABILITIES

Complex survey analysis: HINTS full-sample and replicate-weight designs; NHANES MEC weights, masked strata and PSU-based inference.

Regression & sensitivity analysis: Survey-weighted logistic and linear models, ordinal analysis, interaction testing, model and data-completeness sensitivities.

Repeated cross-sectional analysis: Cross-cycle harmonisation, pooled replicate-weight inference and survey-year interaction testing.

Device-linked behavioural data: Participant-level aggregation of day-level PAM summaries with explicit measurement and completeness caveats.

Evidence synthesis: Two systematic evidence-synthesis projects in progress.

Research design: Formal training through Aberdeen PU5027, in progress.

RELEVANT PROFESSIONAL PRACTICE

Founder, SISFIT Academy

Health and exercise education practice spanning curriculum development, public-facing health communication, knowledge translation and digital content. Historical practice scale includes 300,000+ cumulative cross-platform reach and 3,000+ content assets. Audience/platform metrics are not represented as research participants or validated health outcomes.

KNOWLEDGE TRANSLATION

Experience translating exercise, rehabilitation and health-science concepts into structured curricula, practitioner education and public-facing materials. Professional outputs are distinguished from formal research outputs unless an item meets research-output criteria.

OPEN RESEARCH & RESEARCH INTEGRITY

Reproducible code, analysis plans, machine-generated results, validation reports and interpretation caveats are publicly available for the three secondary-analysis projects. Claims are labelled according to evidence status; cross-sectional designs are not presented as causal.

[Open Research registry](#)